

Part 1: Mathematical Kinematic Derivations

HS-PS2-1

Step 1A (Vessel Position Equation): Initial position $x_0 = +300$ m, constant speed $v = +6.0$ m/s.

$$x_{\text{ship}}(t) = 300 + 6.0 \cdot t$$

Step 1B (Drone Position Equation for $t \geq 20$ s): Launch position $x_0 = 0$ m, speed $v = +18.0$ m/s, delay = 20 s.

Factored: $x_{\text{drone}}(t) = 18.0 \cdot (t - 20)$
Expanded: $x_{\text{drone}}(t) = 18.0 \cdot t - 360$ (for $t \geq 20$ s)

Step 1C (Algebraic Intercept Equating):

$$\begin{aligned} 300 + 6.0 \cdot t &= 18.0 \cdot t - 360 \\ 300 + 360 &= 18.0 \cdot t - 6.0 \cdot t \\ 660 &= 12.0 \cdot t \\ t &= 660 / 12.0 = \mathbf{55.0 \text{ seconds (Rendezvous Time)}} \end{aligned}$$

$$x_{\text{meet}} = 300 + 6.0(55.0) = 300 + 330 = \mathbf{630.0 \text{ meters from dock (Rendezvous Position)}}$$

Check with Drone: $x_{\text{drone}} = 18.0(55 - 20) = 18.0(35) = 630.0 \text{ m} \checkmark$

Part 2: Completed Telemetry Table

Vehicle	t = 0 s	t = 10 s	t = 20 s	t = 30 s	t = 40 s	t = 50 s	t = 55 s (Meet)	t = 60 s
Ship x (m)	300 m	360 m	420 m	480 m	540 m	600 m	630 m	660 m
Drone x (m)	0 m	0 m	0 m	180 m	360 m	540 m	630 m	720 m

Part 4: Mission Engineering Audit & Constraint Solutions

1. Slope Verification: Ship slope = Rise/Run = $(600 - 300)/(50 - 0) = +6.0$ m/s. Drone slope = $(540 - 0)/(50 - 20) = 540/30 = +18.0$ m/s. The visual lines intersect exactly at coordinate **(55 s, 630 m)**, verifying the algebra.
2. Flight Duration vs Clock Time: The mission clock reads $t = 55$ s, but the drone was delayed on dock until $t = 20$ s. Flight duration = $55 \text{ s} - 20 \text{ s} = \mathbf{35.0 \text{ seconds}}$ of actual flight.
3. Battery Safety Audit: Total round-trip distance = 630 m (outbound) + 630 m (inbound) = **1,260 meters**. Since certified battery limit is only 1,200 m, **the drone cannot make it back to dock**. It will exhaust power and ditch in the ocean 60 meters short of the harbor dock.

TEACHER MASTER KEY: CHALLENGE 2

Tactical Highway Surveillance & Patrol Intercept (Piecewise Motion)

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Part 1: Target Piecewise Breakdown

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Leg 1 (0 to 20 s): $v_1 = +15.0 \text{ m/s}$, $\Delta t_1 = 20 \text{ s} \rightarrow \Delta x_1 = (15.0)(20) = +300 \text{ m}$. Coordinate: **(20 s, 300 m)**

Leg 2 (20 to 50 s): $v_2 = 0.0 \text{ m/s}$, $\Delta t_2 = 30 \text{ s} \rightarrow \Delta x_2 = 0 \text{ m}$. Coordinate: **(50 s, 300 m)** (Horizontal flat line)

Leg 3 (50 to 90 s): $v_3 = +25.0 \text{ m/s}$, $\Delta t_3 = 40 \text{ s} \rightarrow \Delta x_3 = (25.0)(40) = +1,000 \text{ m}$.

Final Position: $x_{\text{final}} = 300 + 1,000 = \mathbf{1,300 \text{ meters}}$. Coordinate: **(90 s, 1300 m)**

Part 2: Cruiser Pursuit Strategies

Strategy Alpha (Catch at Rest Stop at $t = 50 \text{ s}$):

Dispatch time: $t = 30 \text{ s}$ | Target arrival time: $t = 50 \text{ s} \rightarrow$ Available travel time: $\Delta t = 50 - 30 = 20.0 \text{ s}$

Required distance: $\Delta x = 300 \text{ m}$

Required Cruiser Speed: $\mathbf{v_{\alpha} = 300 \text{ m} / 20.0 \text{ s} = 15.0 \text{ m/s}}$ (33.6 mph)

Strategy Beta (Catch at Highway End at $t = 90 \text{ s}$):

Dispatch time: $t = 30 \text{ s}$ | Final mission time: $t = 90 \text{ s} \rightarrow$ Available travel time: $\Delta t = 90 - 30 = 60.0 \text{ s}$

Required distance: $\Delta x = 1,300 \text{ m}$

Required Cruiser Speed: $\mathbf{v_{\beta} = 1,300 \text{ m} / 60.0 \text{ s} = 21.67 \text{ m/s} \approx 21.7 \text{ m/s}}$ (48.5 mph)

Part 4: Tactical Analysis & Speed Comparison

1. Horizontal Slope: On an x-t graph, slope = velocity ($\Delta x / \Delta t$). A flat horizontal line has slope = 0, proving instantly that position is unchanging and velocity is zero without needing telemetry.

2. Average Speed vs. Instantaneous Speeds:

Total distance = 1,300 m | Total time = 90.0 s $\rightarrow \mathbf{v_{\text{avg}} = 1,300 \text{ m} / 90 \text{ s} = 14.4 \text{ m/s}}$.

Why lower than 15 m/s and 25 m/s? The 30 seconds of zero velocity at the rest stop significantly drags down the overall time-weighted average speed.

3. Speed Limit Constraint (Cap = 20.0 m/s):

Strategy Alpha requires $15.0 \text{ m/s} \leq 20.0 \text{ m/s}$ (PHYSICALLY POSSIBLE ✓).

Strategy Beta requires $21.7 \text{ m/s} > 20.0 \text{ m/s}$ (EXCEEDS VEHICLE TOP SPEED ✗).

Therefore, **Strategy Alpha is the only viable option** for the patrol interceptor.

TEACHER MASTER KEY: CHALLENGE 3

Part 1: Vector Displacement vs. Scalar Distance

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Step 1A (Pythagorean Vector Magnitude): $\Delta x = 800$ m, $\Delta y = 600$ m.

$$\Delta r = \sqrt{[(800)^2 + (600)^2]} = \sqrt{[640,000 + 360,000]} = \sqrt{[1,000,000]} = \mathbf{1,000 \text{ meters}}$$

Step 1B (Comparison): Rover road distance = 1,800 m. Drone displacement = 1,000 m. The rover travels 800 m more because it is constrained to winding canyon switchbacks that deviate from the direct straight-line vector.

Part 2: Travel Time Calculations

Step 2A (Ground Rover Alpha): Road distance = 1,800 m, speed = 15.0 m/s.

$$t_{\text{rover}} = 1,800 \text{ m} / 15.0 \text{ m/s} = \mathbf{120.0 \text{ seconds (2.0 minutes)}}$$

Step 2B (Rescue Drone Beta Timeline): Cruise speed = 20.0 m/s, flight distance = 1,000 m.

Leg 1 Flight (0 to 500 m): $t_1 = 500 \text{ m} / 20.0 \text{ m/s} = 25.0 \text{ s}$ (Clock: 0 to 25 s)
Thermal Cooldown Pause: $\Delta t_{\text{pause}} = 35.0 \text{ s}$ (Clock: 25 to 60 s, distance remains at 500 m)
Leg 2 Flight (500 to 1,000 m): $t_2 = 500 \text{ m} / 20.0 \text{ m/s} = 25.0 \text{ s}$ (Clock: 60 to 85 s)
Total Drone Arrival Time: $t_{\text{drone}} = 25 + 35 + 25 = \mathbf{85.0 \text{ seconds}}$

Part 4: Mission Debrief & Race Analysis

1. Margin of Victory: Drone arrives at $t = 85$ s; Rover arrives at $t = 120$ s. **Drone Beta wins by 35.0 seconds** ($120 \text{ s} - 85 \text{ s}$).

2. Distance-Time Flat Line Interpretation: From $t = 25$ s to 60 s, the graph is horizontal at $d = 500$ m. The slope (speed) is 0 m/s. The odometer/distance accumulated stops increasing while the drone cools on the ridge.

3. Critical Breakdown Threshold (Honors Proof):

Rover total time = 120 s. Total drone flight time in the air = $25 \text{ s} + 25 \text{ s} = 50 \text{ s}$.

Maximum cooldown duration: $\Delta t_{\text{max_pause}} = 120 \text{ s} - 50 \text{ s} = \mathbf{70.0 \text{ seconds}}$.

If the cooldown exceeds 70 seconds, the rover beats the drone to the climber.